

Ruthenium Brachytherapy for Uveal Melanoma, 1979–2003

Survival and Functional Outcomes in the Swedish Population

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Purpose: To evaluate observed and relative survival rates, enucleation rates, and visual outcome after ruthenium 106 brachytherapy for uveal melanoma.

Design: Retrospective cases series from the Swedish national referral center.

Participants: Five hundred seventy-nine patients (579 eyes) with choroidal or ciliary body melanomas, including 55 tumors more than 7 mm in height, treated with ruthenium episcleral plaques from January, 1979, through April, 2003.

Methods: Clinical and radiotherapy data were extracted from a dedicated database, and survival status was determined through population registries. Tumor size was classified according to the Collaborative Ocular Melanoma Study criteria. The 5- and 10-year relative survival rates were estimated, and univariate and multivariate regression models were constructed for predictive factors on observed survival, enucleation, and visual deterioration.

Main Outcome Measures: Observed and relative survival rate, proportion of secondary enucleation, deterioration of visual acuity to less than 0.5, respectively, to 0.1 or worse.

Results: Tumors were classified as small in 10.5%, medium in 78.4%, and large in 9.2% of patients. The 5- and 10-year observed overall survival rates were 83.3% and 71.5%, respectively, and the corresponding relative rates were 95.5% and 94%, respectively. Factors predicting survival were tumor diameter, patient age, and secondary enucleation. One hundred six patients (18%) underwent enucleation up to 14 years after plaque treatment. The only predictive factor for enucleation was tumor size. At 5 years, 31% of the patients retained 0.5 visual acuity or better, and 49% retained better than 0.1 visual acuity. Predictive factors for visual deterioration were visual acuity and distance from posterior tumor border to the foveola.

Conclusions: After ruthenium brachytherapy for uveal melanoma, the survival rates and visual outcomes in this population-based investigation were similar to previously published results. The eye was retained in 81.7% of patients. Careful patient selection (presently we only treat melanomas 7 mm or smaller in height) and life-long monitoring for recurrences is warranted. *Ophthalmology* 2005;112:834–840 © 2005 by the American Academy of Ophthalmology.

Uveal melanoma threatens both the patient's life and visual function. Treatment methods such as enucleation,^{1,2} brachytherapy,^{3–6} charged particle irradiation,^{7–9} and local resection^{10,11} seem to have similar survival outcomes.^{12–14} Most important is to prevent death resulting from uveal melanoma, but both patients and ophthalmologists also focus on the preservation of visual function, cosmetic appearance, and quality of life.¹⁵

Since Lommatzsch¹⁶ introduced ruthenium brachytherapy for uveal melanoma in the 1960s, many institutions in Europe have gained experience using this isotope for treating small to medium melanomas.^{17–21} In North America, the iodine 125 applicator has become the primary choice for brachytherapy, and this radionuclide therefore was chosen in the brachytherapy arm of the Collaborative Ocular Mel-

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